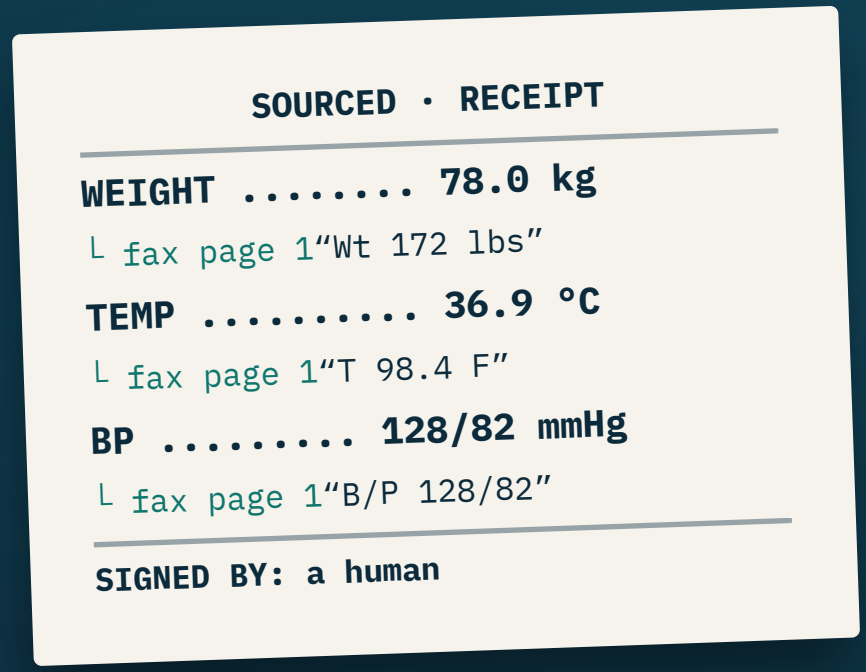
HACKRU FALL 2026 · REGENERON TRACK

Sourced

Clinical data that manages itself,

with receipts



Every trial value is typed twice

290,000

data points in one Phase 3 trial

15–30%
of trial cost is checking

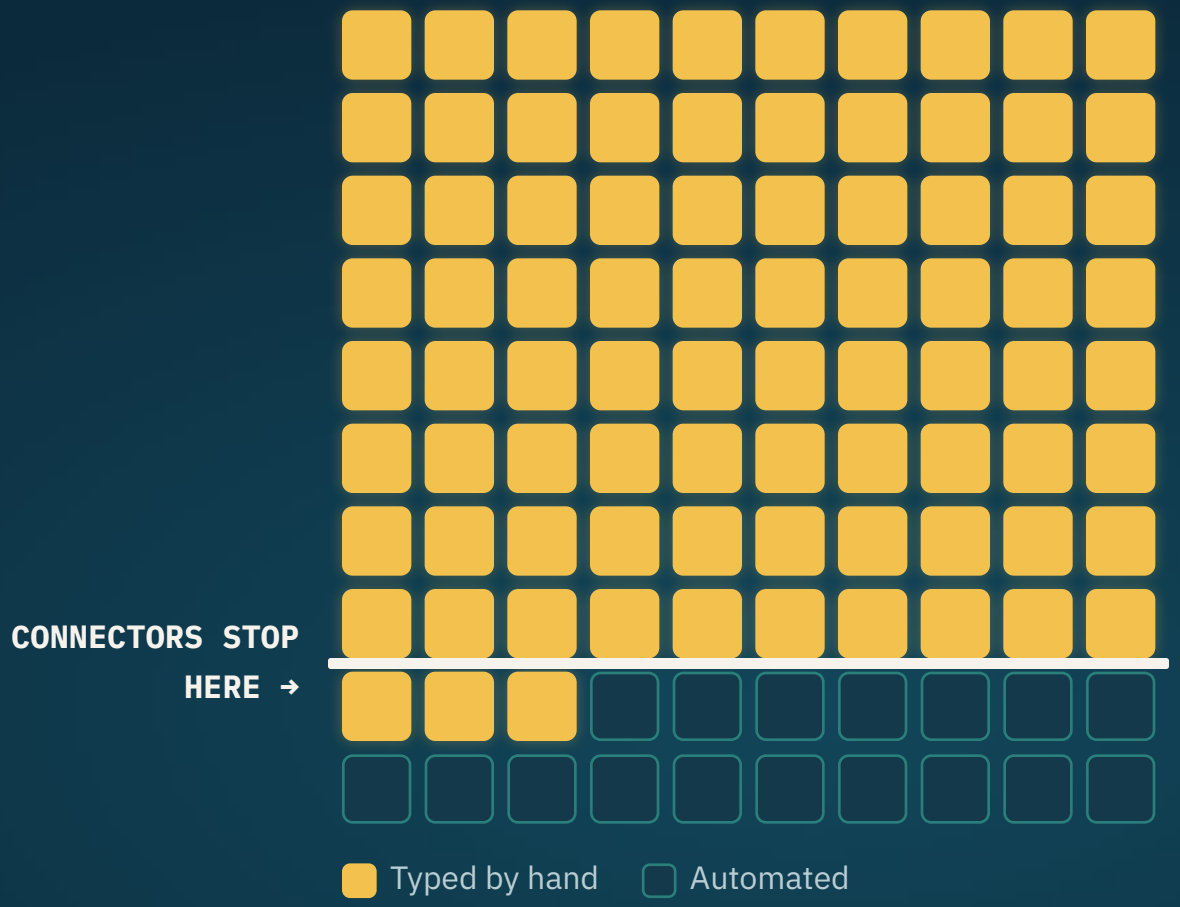
8 days
from visit to data entry

83%

of trial data is still typed by hand

Faxes, scans, clinic notes — beyond what connectors can reach

TransFAIR study · BMJ Health & Care Informatics, 2023

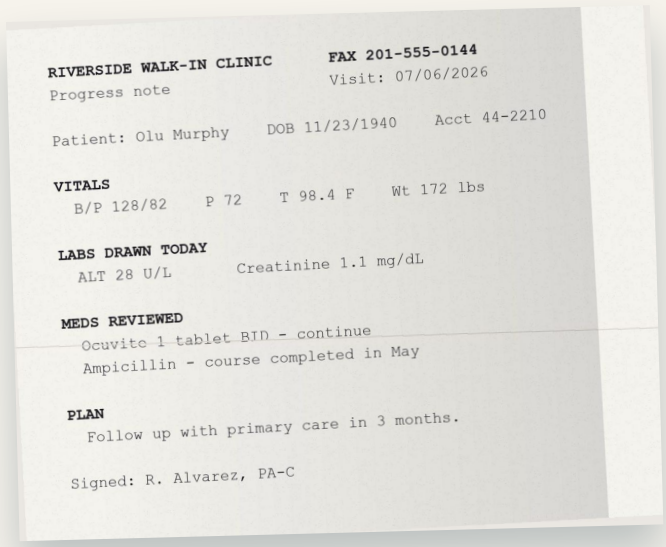


One rule: every value is cited and *traceable*

Sourced reads the other 83% — no receipt, no value

```
weight 78.0 kg ← fax · page 1 · “Wt 172 lbs”
```

A faxed photo → 10 cited facts in 8 seconds



RIVERSIDE WALK-IN CLINIC
FAX 201-555-0144
Progress note
Visit: 07/06/2026

Patient: Olu Murphy DOB 11/23/1940 Acct 44-2210

VITALS
B/P 128/82 P 72 T 98.4 F Wt 172 lbs

LABS DRAWN TODAY
ALT 28 U/L Creatinine 1.1 mg/dL

MEDS REVIEWED
Ocuvite 1 tablet BID - continue
Ampicillin - course completed in May

PLAN
Follow up with primary care in 3 months.

Signed: R. Alvarez, PA-C

Field	On the form	Quote
BP	128/82	“B/P 128/82”
Pulse	72	“P 72”
Temp	36.9 °C	“T 98.4 F”
Weight	78.0 kg	“Wt 172 lbs”
ALT	28 U/L	“ALT 28 U/L”
Ampicillin	Stopped	“completed in May”

Any record to a signed form, in five steps

Follow one value: the weight on the faxed note

1 · AI

Read

Faxes, scans, FHIR become facts with quotes

`“Wt 172 lbs”`

2 · RULES

Store

Every source lands in one patient record

`weight · Jul 6 · fax pg 1`

3 · RULES

Fill

The record fills the trial form, units converted

`Weight: 78.0 kg`

4 · RULES

Verify

Every field re-checked against every source

`Agrees with hospital scale`

5 · HUMAN

Sign

A person resolves flags and exports to the trial database

`Signed · sent to EDC`

Only step 1 uses AI — every step after it checks the one before, so a model error can't reach the form

Flagged and explained — never guessed

⚠ **FLAGGED · Weight**

78.0 kg · Hospital EHR

87.0 kg · Clinic note, page 2

Likely cause: transposed digits

Example flag

Causes it can name

- Unit mix-up (lb vs kg)
- Transposed digits
- Med list missing a stop
- Off the patient's own trend

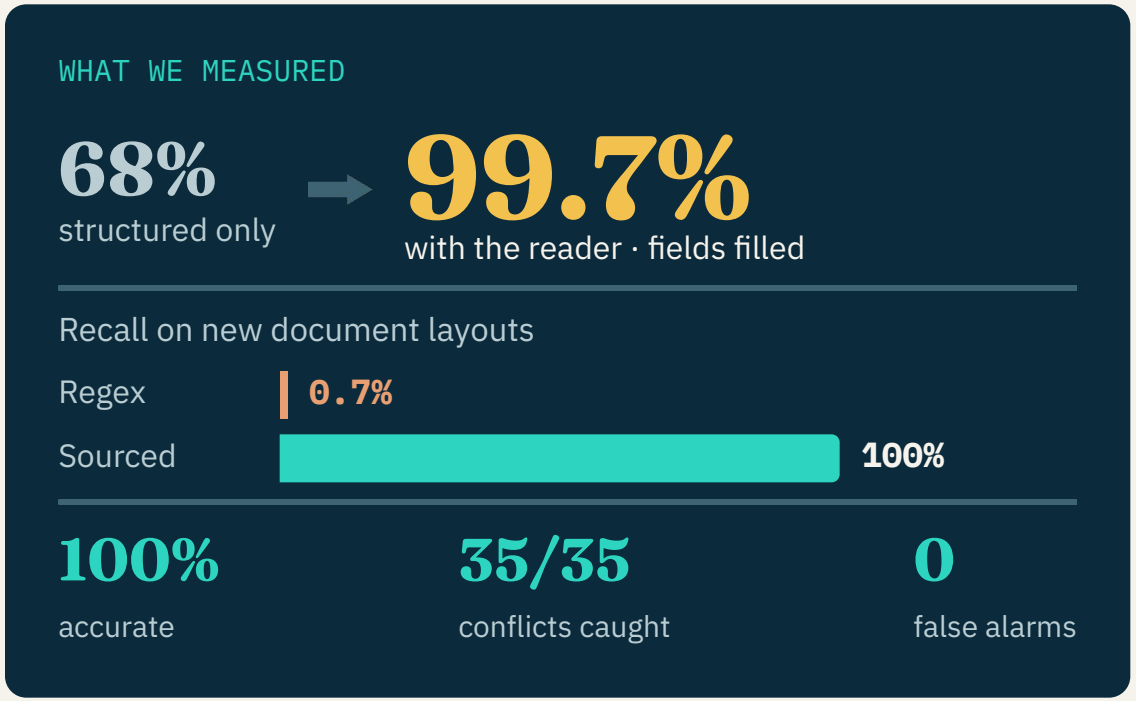
Structured data stalls at ~70% — reading closes the gap

WHAT RESEARCH SHOWS

17% of trial data moved automatically
TransFAIR · BMJ HCI 2023

43% of data points sit in structured EHR
Flatiron & Sanofi · AMIA 2025

~70% of form data a live connector fills
Mount Sinai · 2026



Every receipt answers a regulator

Attributable

ALCOA+ · FDA eSource

Where every value came from

Original

ALCOA+

The exact words, kept verbatim

SOURCED · RECEIPT

VALUE ..... **78.0 kg**

SOURCE ..... fax, page 1

QUOTE ..... “Wt 172 lbs”

CHECKED .... range, units

SIGNED ..... coordinator

HASH ..... 9f3a…c21e

FHIR · LOINC · CDASH · ODM

Accountable

ICH E6(R3)

A person signs, with a reason per flag

Tamper-evident

21 CFR Part 11

Every action hash-chained

The first data agent on your site-agent map

SITE AGENTS OUTLINE

Source-to-EDC

- Source traceable
- Acts once proven
- Coordinator signs

Coordinator

TODAY

Retypes every visit, days later

WITH SOURCED

Reviews only the flags

Monitor

TODAY

Travels on site to compare field by field

WITH SOURCED

Verifies with one click

Data manager

TODAY

Raises queries months later

WITH SOURCED

Conflicts caught at fill time

One Postgres holds the whole ledger

Hypertable

Every fact over time, per patient

```
chunked by observed_at
```

Compression

Old history shrinks, still queryable

22 / 24

chunks compressed

Aggregates

Live chart, refreshed incrementally

```
time_bucket('1 day')
```

pgvector

Find the fax that mentions a drug

```
384-dim HNSW index
```

RUNS ON `Tiger Cloud` `PostgreSQL 18` `TimescaleDB 2.30` `pgvector 0.8`

From proof to a site's daily tool

01

Fax & EHR intake

Documents flow in on their own

02

Live EDC export

Signed forms land in the trial database

03

Scans at volume

A site's whole paper backlog

04

Acts alone once proven

Stops asking about fields it gets right

The model reads
The rules fill and check
The human signs

Sourced github.com/AyaanFaisal21/HackRUF26

